

Kundai Farai Sachikonye

Zeltnerstrasse 30
90443, Nuernberg, Germany
☎ (+49) 1626386344
✉ kundai.sachikonye@bitspark.com
🌐 github.com/fullscreen-triangle



Computational scientist: digital image analysis of microscopy data, neural-circuit / connectome modelling, and transcriptome analysis.

Profile

Computational researcher focused on **quantitative analysis of imaging data, neural-circuit / connectome modelling**, and **transcriptome analysis**. My background is computational rather than wet-lab Drosophila genetics or laser-microscope operation; I would contribute to the AG Neurobiologie's complementary computational and image-analysis work.

Selected Computational Work

- Microscopy image analysis **Quantitative image-analysis framework** — segmentation, PSF deconvolution, scale-field / coordinate reconstruction, and morphometry with uncertainty quantification, for fluorescence and phase-contrast microscopy. GPU-accelerated; live tool. hieronymus-omega.vercel.app
- Computational neuroscience **Neural-circuit modelling** (project "Olduvai") — closed-circuit decomposition of motor/neural signals (supraspinal vs. peripheral), applied to neurodegeneration (Parkinson's, ataxia) from physiological time-series; circuit-graph (connectome-style) network analysis.
- Transcriptome / multi-omics **gospel** — integration of transcriptomics with genotype and metagenomics; validated against 1000 Genomes, gnomAD, UK Biobank. github.com/fullscreen-triangle/gospel

Experience

- 2021–present **Independent Computational Researcher, Bitspark GmbH**
Full-time computational research: quantitative image analysis, neural-circuit / connectome modelling, transcriptome and multi-omics analysis. Scripted, reproducible pipelines with statistical evaluation; open, documented, deployed tools.
- 2019–2021 **Computational Lipidomics — Bioinformatics, HPC & Teaching, Technische Universität München, Munich**
High-throughput data analysis at HPC scale; statistical evaluation of large datasets. Taught at Master's level and supervised student work.
- 2017–2018 **Glycomics and Proteomics, Universitätsklinikum Hamburg-Eppendorf, Hamburg**
LC-MS/MS and MALDI-TOF data pipelines; reproducible high-throughput workflows.
- 2013 **Cancer Biomarker Discovery, University Medical Center Groningen, Groningen**
UPLC-MS/MS proteomics; statistical analysis of high-dimensional data.

Education

- 2019–2021 **Doctoral research, Computational Lipidomics, Technische Universität München**
(Doctoral research; not completed.) Computational analysis of high-throughput data.

- 2018–2019 **MSc Computational Biology**, *Universität Tübingen*
Sequence/transcriptome analysis, statistics, ML for biological data.
- 2014–2017 **MSc Pharmaceutical Biotechnology**, *HAW Hamburg*
- 2011–2014 **BSc Biochemistry and Cell Biology**, *Jacobs University Bremen*
Thesis: DNA interstrand crosslinking — mammalian cell culture and fluorescence microscopy.

Technical Skills

- Image analysis Digital image analysis (segmentation, deconvolution, registration, morphometry); statistical evaluation of imaging data; GPU/GLSL image processing
- Computational neuroscience Neural-circuit and connectome-style network/graph analysis; physiological time-series modelling
- Data analysis Transcriptome analysis; statistics; scripted, reproducible pipelines (Python, R); HPC (Ray, Dask, SLURM)
- Languages (programming) Python (primary), Rust, R, Bash, SQL

Teaching & Languages

- Teaching Taught at Master's level at TUM (2019–2021); supervised student work.
- English Native / Fluent (C2) *lab language*
- German Conversational *resident in Germany since 2011*